

Generation Interconnection System Impact Study Report

For Queue Project Q21-1848

Archie 138 kV
October 27, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1848. The Interconnection Customer proposes a 115.7 MW solar photovoltaic facility interconnecting at the Archie 138 kV substation in the Pennsylvania area. The request was received on August 05, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 609 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,884,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Whillock – Boyntonville 138kV	Las Cruces 3602 – Cadillac Renewable Energy 138kV	135.1	256	18.0
Walmart – Keswick 138kV line	Medora – P.T.S 345kV	121.8	242	21.6
Keswick – Arsenal Hill 138kV L	Medora – P.T.S 345kV L	118.6	268	21.6
Tecuya – Whillock 138kV	Boyntonville – Whillock 138kV	117.7	201	19.2
Arsenal Hill – Forbing Rd Substation 138kV Line	Medora – P.T.S 345kV	116.6	288	21.6
Las Cruces 3602 – Cadillac Renewable Energy 138kV	Whillock – Boyntonville Substation 138kV	115.1	250	19.2
East Vicksburg – Las Cruces 3602 138kV	Whillock – Boyntonville 138kV	114.9	235	19.2
Whillock – Forbing Rd 138kV	P.T.S – Medora 345kV	114.7	309	21.6
Tecuya – E Vicksburg 138kV	Whillock – Boyntonville 138kV L	114.5	223	19.2
Pulliam Sw Yd Substation – Southside Generating Station Substation 138kV Line	Forbing Rd – Whillock 138kV	108.9	265	7.0
Pulliam Sw Yd – Shepherds Central 138kV	Forbing Road – Whillock 138kV Line	106.5	260	7.0
Shepherds Central – Woodville 138kV line	Whillock – Forbing Rd 138kV	103.2	257	7.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Whillock – Boyntonville 138kV	Rebuild line with higher-capacity conductor	\$1,409,000	21% (\$292,000)
Walmart – Keswick 138kV	Rebuild line with higher-capacity conductor	\$1,347,000	23% (\$305,000)
Keswick – Arsenal Hill 138kV L	Rebuild line with higher-capacity conductor	\$1,621,000	23% (\$367,000)
Tecuya – Whillock 138kV	Rebuild line with higher-capacity conductor	\$2,142,000	27% (\$577,000)
Arsenal Hill – Forbing Rd Substation 138kV Line	Rebuild line with higher-capacity conductor	\$1,474,000	23% (\$333,000)
Las Cruces 3602 Substation – Cadillac Renewable Energy 138kV Line	Rebuild line with higher-capacity conductor	\$5,476,000	27% (\$1,475,000)
Las Cruces 3602 – East Vicksburg 138kV	Rebuild line with higher-capacity conductor	\$441,000	27% (\$119,000)
Whillock – Forbing Rd 138kV	Rebuild line with higher-capacity conductor	\$4,311,000	23% (\$975,000)
Tecuya – E Vicksburg 138kV	Rebuild line with higher-capacity conductor	\$589,000	27% (\$159,000)
Southside Generating Sta – Pulliam Sw Yd 138kV	Reconductor line	\$451,000	18% (\$80,000)
Shepherds Cetral – Pulliam Sw Yd Substation 138kV L	Reconductor line	\$793,000	18% (\$141,000)
Shepherds Central – Woodville 138kV	Reconductor line	\$344,000	18% (\$61,000)
	Total allocated to project		\$4,884,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.